Fiscal Year:	FY 2013	**Task Last Updated:**	FY 05/09/2013
PI Name:	Poirier, David Sc.D.		
Project Title:	Research for ESA Space Experiments on Directional Solidification of Alloys		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	poirierd@email.arizona.edu	**Fax:**	FY 520-621-8059
PI Organization Type:	UNIVERSITY	**Phone:**	520-621-6072
Organization Name:	The University of Arizona		
PI Address 1:	1235 East James E Rogers Way		
PI Address 2:	Department of Materials Science and Engineering		
PI Web Page:	http://www.mse.arizona.edu/		
City:	Tucson	**State:**	AZ
Zip Code:	85721-0012	**Congressional District:**	7
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	OTHER
Start Date:	05/15/2008	**End Date:**	06/30/2014
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	3
No. of Master's Candidates:	4	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Downey, James1 Patton	**Contact Phone:**	
Contact Email:	patton.downey@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS Expedition 19/20 NOTE: Extended to 6/30/2014 per NSSC information (Ed., 4/9/14) NOTE: Extended to 5/14/2014 per PI and NSSC information (Ed., 4/5/2013) NOTE: End date extended to 5/14/2013 (from 5/14/2011) in Dec 2010 per NSSC information (Ed., 2/14/2011)		
Key Personnel Changes/Previous PI:	Professor Olaya Florez of the Universidad de Colombia is with the research team at The University of Arizona while he is on sabbatical for most of 2013.		
COI Name (Institution):	Erdmann, Robert (University of Arizona) Tewari, Surendra (Cleveland State University)		
Grant/Contract No.:	NNX08AN49G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	In this research program two experiments were processed on the International Space Station (ISS) to study the effect of microgravity on directionally solidified alloy microstructures and segregation. At Cleveland State University (CSU) terrestrial directional solidification experiments on Al-7%Si alloys were done using different growth conditions to effect variations in dendritic metrics. Numerical modeling to simulate the transport phenomena and solidification of the planned experiments is progressing at the University of Arizona (UA). Based on previous research, it is clear that thermosolutal convection, which always occurs when alloys solidify on earth, masks the effects of diffusional transport during solidification. With terrestrial experiments, diffusive and convective phenomena occur simultaneously, which complicate the study of dendritic growth. In microgravity, however, directional solidification occurs with no thermosolutal convection, so the convection and attendant macrosegregation are almost completely eliminated, and the growth of the cast dendritic- microstructure is controlled by a diffusional processes. We can also compare segregation and microstructure in specimens terrestrially solidified and those solidified in microgravity. The technological significance of the research is that it provides scientific underpinning to our understanding and control of how single-crystal castings solidify. Single-crystal castings are critical components in high-temperature gas-turbine engines that are used in high-speed aircraft and land-based power generators. The research is being done by the team at The University of Arizona and Cleveland State University. We are part of MICAST (Microstructure Formation in Casting of Technical Alloys) led by ESA-researchers in the German Aerospace Center in Germany. One sample was processed in the ISS in August, 2010 and a second was processed on the ISS in Feb. 20011. Both samples have been returned to Earth and have been under study at CSU and MSFC. Computer simulations have been ongoing at UA.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research is relevant to a process known as directional solidification, which is used for producing critical castings that are used in gas turbine engines. Gas turbine engines power aircraft (i.e., often called "jet engines") and are also used to generate electric power. These castings are made of vacuum melted alloys that are capable of withstanding the demanding conditions at high temperatures and consequently the metallurgical structure of the alloys must be of the highest quality. The research is leading to an improved understanding of how microstructures and segregation form in directionally solidified castings; hence the science of producing such castings will be enhanced as a result of the research. In addition, the research findings relate to many other types of castings, including billets and ingots, of alloys (i.e., steel, aluminum alloys, etc.) that are important in the "manufacturing chain."
Task Progress:	This report is mainly on the results of examining the structure and segregation of the two samples of hypoeutectic Al-7Si alloy that were directionally solidified on the International Space Station (ISS). The two samples are identified as MICAST6 and MICAST7. Dendritic arrays of the entire cross sections were composed of approximately thirty-five images, from which seamless composite images were made by a computerized method established in the program. Such images from specimens taken from the research samples were used to measure two dendritic metrics: average primary dendrite arm spacing and the primary dendrite trunk diameter. We have seen that the space grown samples follow the predictions of the dendritic growth models, whereas the metrics of the terrestrially solidified samples are significantly affected by thermosolutal convection driven by buoyancy in 1-g. Using software developed in the grant-program, the output from twelve thermocouples attached to the space experiments was reduced to graphical form to show (i) temperature vs. position along the solidifying sample at a given time, (ii) the thermal gradients and (iii) the solidification rates. Electron back scattered diffraction was employed to assess the uniformity of the dendritic orientations. Unexpectedly, detached tertiary arms were observed in the dendritic arrays. The mechanism leading to the detachment is from the ripening phenomena, driven by the capillarity effects in the mushy zone. Processing by directional solidification (DS), required that the samples were maintained in a stationary thermal field in order to melt back to a seed crystal and to establish an all-liquid zone for the DS. During this holding period, there is diffusion of the solute (Si) in the interdendritc liquid of the mushy zone toward the all-liquid zone. The dendrite tips in the leading part of the mushy zone become blunted, and at the eutectic isotherm a band of the Al-rich phase is established. This is studied by microstructure examinations and computer modeling.
Bibliography Type:	Description: (Last Updated: 06/06/2025)
Articles in Peer-reviewed Journals	Van Hoose JR, Grugel RN, Tewari SN, Erdmann R, Poirier DR. "Observation of misoriented tertiary dendrite arms during controlled directional solidification in aluminum-7wt pct silicon alloys." Metallurgical and Materials Transactions A. 2012 Dec;43A(12):4724-31. http://dx.doi.org/10.1007/s11661-012-1260-z , Dec-2012
Dissertations and Theses	Rajamure RS. "Directional Solidifcation of Al-7 wt% Si Alloy." Thesis, Cleveland State University, June 2010. , Jun-2010
Dissertations and Theses	Pakiru MS. "Effect of step change in growth speed during directional solidifcation on array morphology of Al-7 wt% Si alloy." Thesis, Cleveland State University, June 2011. , Jun-2011
Dissertations and Theses	Panova O. "Automated Analysis of Dendritic Structures in Binary Alloys." Thesis, The University of Arizona, Tucson, August 2012. , Aug-2012